Lecture 21 (More MIP Examples)

April 2, 2026

Today we will give a few examples of MIP problems. We will formulate a few problems as MIPs as well.

Examples

Ex 1: Another Scheduling Example

Statement

Suppose a factory has to create N vehicles. Each vehicle can be produced on one of M production lines. Suppose these are $m_1, m_2, \dots m_M$. Suppose m_i costs c_i to build one vehicle and suppose that this line will take t_i hours to produce a vehicle. Suppose the factory must create the vehicles within T hours. Find the lowest cost schedule such that N vehicles are produced within time T.

Formulation

First, we should find the decision variables. We can choose how many vehicles to produce on each line. This yields the following integer variables:

$$x_i \in \mathbb{Z}_{\geq 0}$$

where x_i represents the number of vehicles produced on production line m_i. Now, for the constraints. The first constraint ensures that all of the vehicles are produced on time.

$$t_i x_i \leq T \ \forall i$$

We are assuming each production line can run in parallel, so each production line (assuming it can run continuously through the time frame) will run for $t_i x_i$ hours to run. We also need to ensure the total number of vehicles produced is correct:

$$\sum_{i=1}^{M} x_i = N$$

Now, for the objective:

$$\text{minimize} \sum_{i=1}^{M} c_i x_i$$

In a more familiar form, this is stated as

$$\text{minimize} \sum_{i=1}^{M} c_i x_i$$

s.t.

$$t_i x_i \leq T \ \forall i$$

$$\sum_{i=1}^{M} x_i = N$$

where

$$x_i \in \mathbb{Z}_{\geq 0}$$

Ex 2: Cutting Problem

Statement

Suppose a factory can buy metal rods from a collection with corresponding lengths $L_1, L_2, \ldots, L_n$ and with corresponding costs, $c_1, c_2, \ldots, c_n$. The factory needs to have rods of lengths $l_1, l_2, \ldots, l_m$. Suppose the rods the factory buys can be cut into smaller lengths but cannot be joined into longer lengths. Find the collection of rods that minimizes costs while satisfying the factory demands (assuming the appropriate cuts are made).

Formulation

First, we should decide on the decision variables. We can decide to cut rod i to get a rod length l_j. We also could decide whether or not to purchase a rod. This yields two classes of decision variables:

$$x_{ij} = \begin{cases} 1 & \text{If rod } L_i \text{ is cut to form rod } l_j \\ 0 & \text{Otherwise} \end{cases}$$

$$y_i = \begin{cases} 1 & \text{If rod } L_i \text{ is purchased} \\ 0 & \text{Otherwise} \end{cases}$$

Now for the constraint. We need to make sure that each rod that is cut is purchased. To do so, we will use $M \gg 0$ (similar to how we did for big M).

$$\sum_{j=1}^{m} x_{ij} \leq M y_i \ \forall i$$

If any $x_{ij} = 1$, then notice this forces $y_i = 1$. For each L_i, we need to ensure the cut rod lengths do not exceed the original rod's length.

$$\sum_{j=1}^{m} x_{ij} \leq L_i \ \forall i$$

We also need to make sure the factory's demands are met.

$$\sum_{i=1}^{n} x_{ij} = 1 \ \forall j$$

Finally, we can define our objective function:

$$\text{minimize} \ \sum_{i=1}^{n} c_i y_i$$

If we write this in a more familiar form, we see:

$$\text{minimize} \ \sum_{i=1}^{n} c_i y_i$$

$$\sum_{j=1}^{m} x_{ij} \leq M y_i \ \forall i$$

$$\sum_{j=1}^{m} x_{ij} \leq L_i \ \forall i$$

$$\sum_{i=1}^{n} x_{ij} = 1 \ \forall j$$

$$x_{ij}, y_i \in \{0, 1\}$$

Ex 3: Selling Jam

Let's give an example of a MIP that isn't a BIP or an IP.

Statement

Suppose you are selling jam at various farmer's markets. You have produced N gallons of jam. You can sell your jam at one of M markets. Market i will yield a profit of p_i dollars for each gallon of jam sold. Furthermore, market i will have a one time vendor registration cost of c_i that must be paid in order to sell any jam at the market. Market i also has a sales capacity of u_i gallons of jam. This is the most jam that can be sold at market i. Any jam that is not sold will cost you a dollars per gallon. What sales plan maximizes your profit?

Formulation

First, let's determine the decision variables. We need to determine how much jam to sell at each market. If we assume that jam is a continuous variable, then we have

$$x_i \in \mathbb{R}_{\geq 0}$$

is the number of gallons of jam sold at market i. We also have a variable

$$b \in \mathbb{R}_{\geq 0}$$

to represent the number of unsold gallons of jam. Finally, we have

$$y_i = \begin{cases} 1 & \text{If any jam is sold at market } i \\ 0 & \text{Otherwise} \end{cases}$$

Now, for the constraints. First, we need to ensure that all jam is accounted for (conservation of jam)

$$b + \sum_{i=1}^{M} x_i = N$$

We also need to ensure that the market capacities are satisfied:

$$x_i \leq u_i \ \forall i$$

and finally, we need to ensure the vendor registration cost is paid for all markets. We will use the same M trick as seen in example 2. For a tighter bound, we could use N from the problem statement for M. This is because we already have an upper bound on the jam from conservation of jam.

$$\sum_{i=1}^{M} x_i \leq N y_i \ \forall i$$

Now, we need to define the objective function:

$$\text{maximize } \sum_{i=1}^{M} p_i x_i - ab - \sum_{i=1}^{M} c_i y_i$$

This yields the more familiar formulation below:

$$\text{maximize } \sum_{i=1}^{M} (p_i x_i - c_i y_i) - ab$$

s.t.

$$b + \sum_{i=1}^{M} x_i = N$$

$$x_i \leq u_i \ \forall i$$

$$\sum_{i=1}^{M} x_i \leq N y_i \ \forall i$$

$$x_i, b \in \mathbb{R}_{\geq 0}, y_i \in \{0, 1\}$$